

ALM-ILS 2026 Satellite Workshop

Bench to Biology: Planning and Running Human Lipidomics and Lipoprotein Studies

18th October 2026 | 9:00 am - 1:30 pm | FREE | Limited to 10 places only

Australian National Phenome Centre (ANPC), Harry Perkins Research Institute (South),
5 Robin Warren Drive, Murdoch, Perth WA

About the Workshop

Quantitative lipoprotein profiling by NMR and lipidome characterisation by LC-MS are increasingly being incorporated into human cohort studies, clinical trials, and epidemiological research. Incorporating these platforms into a study involves a chain of decisions, in study design, sample handling, QC, and data processing, each of which influences the quality and interpretability of the final dataset.

This workshop is designed for researchers who are considering adding lipoprotein or lipidomics analysis to a human study, or who want a clearer understanding of what these platforms can deliver and what it takes to generate high-quality, interpretable data. Delivered at the Australian National Phenome Centre (ANPC), a national resource in large-scale phenotyping, the programme will walk participants through that entire chain, from choosing the right platform and collecting samples correctly, through to QC, data processing, and translating outputs into biologically and clinically meaningful findings.

A guided facility tour and morning coffee break provide an opportunity to see the instrumentation in person and speak informally with ANPC researchers.

Who Should Apply

This workshop is aimed at PhD students and early to mid-career researchers (within 10 years of PhD completion, *note - researchers outside of this scope will still be considered subject to availability*) in human health, clinical, or epidemiological research who are considering incorporating lipoprotein profiling or lipidomics into their work, or who want to deepen their understanding of how to design and execute these studies rigorously. No prior hands-on experience with NMR or LC-MS is required.

Details:

Date	Satellite to ALM-ILS 2026, October 2026
Time	9:00 am – 1:30 pm
Location	Australian National Phenome Centre, Harry Perkins Research Institute (South), 5 Robin Warren Drive, Murdoch, Perth WA. Short 5-minute walk from Murdoch Train Station. Free weekend parking available on Murdoch University Campus.
Capacity	10 places
Eligibility	Current postgraduate students or EMCR researchers within 10 years of PhD completion (<i>please note researchers outside of this scope will still be considered subject to availability, please contact admin@alm-ils-2026.org</i>)
Cost	Free

How to Register

Places are limited to 10 participants and will be allocated competitively. Priority will be given to current postgraduate students and academic researchers within 15 years of PhD completion (consistent with the Australian Academy of Science EMCR definition). Researchers outside this scope are welcome to apply and will be considered subject to availability.

Applications are submitted through the registration portal for the main ALM-ILS 2026 meeting (www.alm-ils-2026.org). Applications close 18 September 2026.

Workshop Facilitators



Dr. Philipp Nitschke is the NMR Lead at the ANPC, specialising in the development and application of advanced nuclear magnetic resonance spectroscopy methods for high-throughput metabolic phenotyping. His research heavily focuses on quantitative lipoprotein, lipidomic, and metabolite profiling, contributing significantly to the discovery of systemic molecular biomarkers for complex diseases and inflammatory responses.



Dr. Reika Masuda is the lead bioinformatician at the ANPC, specialising in the application of advanced computational modelling and machine learning to large-scale metabolic and lipidomic datasets. Her research focuses on integrating complex omics data to identify systemic biomarkers for cardiovascular risk, infectious diseases, and inflammatory responses in human populations.



Dr. Luke Whiley is a Senior Lecturer in the School of Diagnostic and Therapeutic Sciences at Curtin University and leads the Lipid and Metabolic Phenotype Group within the Curtin Medical Research Institute. He holds an adjunct appointment at the ANPC and has extensive experience in targeted and untargeted LC-MS lipidomics across neurodegeneration, inflammatory injury, and large human cohort studies.

Detailed programme

SESSION 1 - Platforms, Study Design, and Sample Collection

9:00 - 9:10 | Welcome and Overview

Introduction to the workshop, the ANPC, and the facilitators. Overview of the day's programme and objectives.

9:10 – 9:35 | What Do These Platforms Actually Measure?

A combined, introduction to both platforms - what NMR lipoprotein profiling and LC-MS lipidomics each measure, what biological questions they are best suited to address, and how they are complementary. Includes a brief orientation to key outputs: lipoprotein subclass quantification and inflammatory markers and lipid species and class profiles. Framed around the question: *which platform, or combination, is right for your lipid study?*

9:35 – 10:00 | Study Design Considerations

What you need to think through before a single sample is collected. Topics include:

- Defining the analytical question and selecting the appropriate matrix (plasma vs serum; fasting vs non-fasting)
- Sample size and statistical power in the context of multi-lipid datasets

- Cross-sectional vs longitudinal designs: implications for sample management and QC
- Integrating lipidomics or lipoprotein analysis into an existing study vs designing around it from the outset

10:00 – 10:25 | Sample Collection and Handling

The decisions made at the point of collection can have a disproportionate impact on data quality. This segment covers:

- Blood collection tubes, anticoagulants, and their effects on NMR and LC-MS measurements
- Processing protocols: centrifugation timing, temperature, aliquot volume
- Pre-analytical variables: fasting duration, time-of-day effects, haemolysis, lipaemia
- Freeze-thaw cycles, long-term biobanking, and sample stability
- Practical considerations when working with biobanked samples from existing cohorts

10:25 – 10:50 | Running Large Human Studies: Logistics and Planning

Moving from a pilot to a large cohort study introduces logistical complexity that is easy to underestimate. Topics include:

- Batch planning and sample randomisation to minimise systematic bias
- Sample tracking, chain of custody, and laboratory information management
- Working with a core facility: what to provide, what to expect, and how to plan timelines
- Budgeting and costing for large lipidomics and lipoprotein studies

10:50 – 11:00 | Discussion and open questions to close Session 1.

11:00 – 11:30 | LAB TOUR AND MORNING COFFEE

A guided tour of the ANPC facility, including high-field NMR spectrometers, LC-MS platforms, and sample preparation and data processing infrastructure. An informal opportunity to speak with ANPC researchers and technical staff over coffee.

SESSION 2 –Data pre-processing, review and quality assessment. Biological Interpretation

11:30 – 12:20 | Data Pre-Processing: From Raw Output to Analysis-Ready Data

A practical walkthrough of the QC and processing steps between instrument output and a clean, analysis-ready dataset. Covering both platforms, topics include:

- NMR: spectral processing, deconvolution, and quantification; navigating Bruker IVDR reports and understanding what each output represents
- LC-MS: peak detection, lipid annotation, adduct interpretation, and common annotation pitfalls
- Quality control strategies, e.g. pooled QC samples preparation, placement, and what they tell you
- Quality control strategies to ensure robustness of data
- Run-order effects and signal drift: detection and correction

12:30 – 12:50 | Things to Look Out For: Common Pitfalls and How to Catch Them

Drawn from experience running large-scale human studies, this segment covers the problems that most commonly affect data quality and interpretability:

- Batch effects: causes, detection, and correction approaches

- Sample outliers: haemolysis, lipaemia, sample mix-up, and extraction failures
- Red flags in a lipidomics or lipoprotein dataset that are worth investigating before analysis

12:50 – 13:15 | From Data to Biology: Interpretation and Reporting

The final analytical step: translating processed data into findings that are biologically and clinically meaningful:

- Reading NMR lipoprotein subclass outputs and IVDR-derived cardiometabolic markers in a clinical or epidemiological context
- Summarising lipidome data: lipid class and subclass strategies, saturation indices, and rationale-driven feature selection
- Visualisation approaches for lipoprotein and lipidomics data
- Reporting standards and what journals and reviewers expect

13:15 – 13:30 | Open Discussion: Designing Your Study, questions, take home tops to close and Wrap-Up

An open, facilitated discussion with both workshop leaders. Participants are encouraged to bring specific questions about their own research projects or planned studies. Summary of key takeaways, overview of ANPC services and contact information, and further resources.
